

Question 1

1. Consider the observation model $x_n = \log \theta + w_n$ $n = 1, \dots, N$, where $w_n \sim \mathcal{N}(0, 1)$ are i.i.d. and $\theta > 0$ is an unknown deterministic parameter.

a) Find Estimator

- a. Propose an estimator of θ from the observations $\mathbf{x}^{(N)} = [x_1, \dots, x_N]^T$.

Since θ is a deterministic parameter we will prefer using ML estimator.

- We will derive the expression of the ML estimator by definition :

$$\hat{\theta}_{ML}(\underline{x}) = \underset{\theta \in \Omega_\theta}{\operatorname{argmax}} \{f_{\underline{x}|\theta}(\underline{x}|\theta)\}$$

- For that purpose, we will find the conditional distribution of $\underline{x}|\theta$:

$$x_n = \log \theta + w_n : w_n \stackrel{i.i.d}{\sim} \mathcal{N}(0, 1)$$

$$\underline{x}^{(N)} \sim \mathcal{N}(\mathbf{1}_N \cdot \log \theta, I_N) : \mathbf{1}_N = \begin{pmatrix} 1 \\ 1 \\ \vdots \\ 1 \end{pmatrix} \in \mathbb{R}^N$$

- We can derive the conditional joint PDF of the samples :

$$f_{\underline{x}|\theta}(\underline{x}; \theta) = \prod_{i=1}^N f_{x_i|\theta}(x_i; \theta) = \prod_{i=1}^N \frac{1}{\sqrt{2\pi}} \cdot e^{-\frac{(x_i - \log \theta)^2}{2}} = (2\pi)^{-\frac{N}{2}} \cdot \exp \left[-\frac{1}{2} \sum_{i=1}^N (x_i - \log \theta)^2 \right]$$

- We notice that for convenience, we can find the max argument of the $\ln$ of the conditional joint distribution without changing our goal :

$$\hat{\theta}_{ML}(\underline{x}) = \underset{\theta \in \Omega_\theta}{\operatorname{argmax}} \{f_{\underline{x}|\theta}(\underline{x}|\theta)\} = \underset{\theta \in \Omega_\theta}{\operatorname{argmax}} \{\ln f_{\underline{x}|\theta}(\underline{x}|\theta)\}$$

$$\dots = \ln \left[(2\pi)^{-\frac{N}{2}} \cdot \exp \left[-\frac{1}{2} \sum_{i=1}^N (x_i - \log \theta)^2 \right] \right] = -\frac{N}{2} \ln(2\pi) + \left[-\frac{1}{2} \sum_{i=1}^N (x_i - \log \theta)^2 \right]$$

- We will find the max argument by taking the derivative and comparing to 0 :

$$\xrightarrow{\frac{d}{d\theta}} -\cancel{\frac{1}{2}} \sum_{i=1}^N \cancel{2} \cdot (x_i - \log \theta) \cdot \left(-\frac{1}{\theta}\right) = \cancel{\frac{1}{\theta}} \left[N \cdot \log \theta - \sum_{i=1}^N x_i \right] = 0$$

$$\theta > 0 \implies N \log \hat{\theta} = \sum_{i=1}^N x_i \implies \hat{\theta} = \exp \left[\frac{1}{N} \cdot \sum_{i=1}^N \right]$$

✓ Result

We get the final derivation of the MLE :

$$\hat{\theta}_{ML}(\underline{x}^{(N)}) = \exp \left[\frac{1}{N} \cdot \sum_{i=1}^N \right] = \exp(\bar{x})$$

b) Analyzing The Results

- b. Using MATLAB/ Python, evaluate the bias, variance, and MSE of the estimates for different values of N and θ as follows. Let $\hat{\theta}_N = \hat{\theta}(\mathbf{x}^{(N)})$ denote the proposed estimator and define the estimation error for the i^{th} realization of $\mathbf{x}^{(N)}$ as $\varepsilon_N(\mathbf{x}_i^{(N)}, \theta) = \hat{\theta}(\mathbf{x}_i^{(N)}) - \theta$ where $i = 1, \dots, L$ stands for an index of the i^{th} realization of $\mathbf{x}^{(N)}$. The bias, variance, and MSE of the estimates are given by

$$\text{bias}(N, \theta) = \frac{1}{L} \sum_{i=1}^L \varepsilon_N(\mathbf{x}_i^{(N)}, \theta), \quad \text{var}(N, \theta) = \frac{1}{L} \sum_{i=1}^L \left(\varepsilon_N(\mathbf{x}_i^{(N)}, \theta) - \text{bias}(N, \theta) \right)^2$$

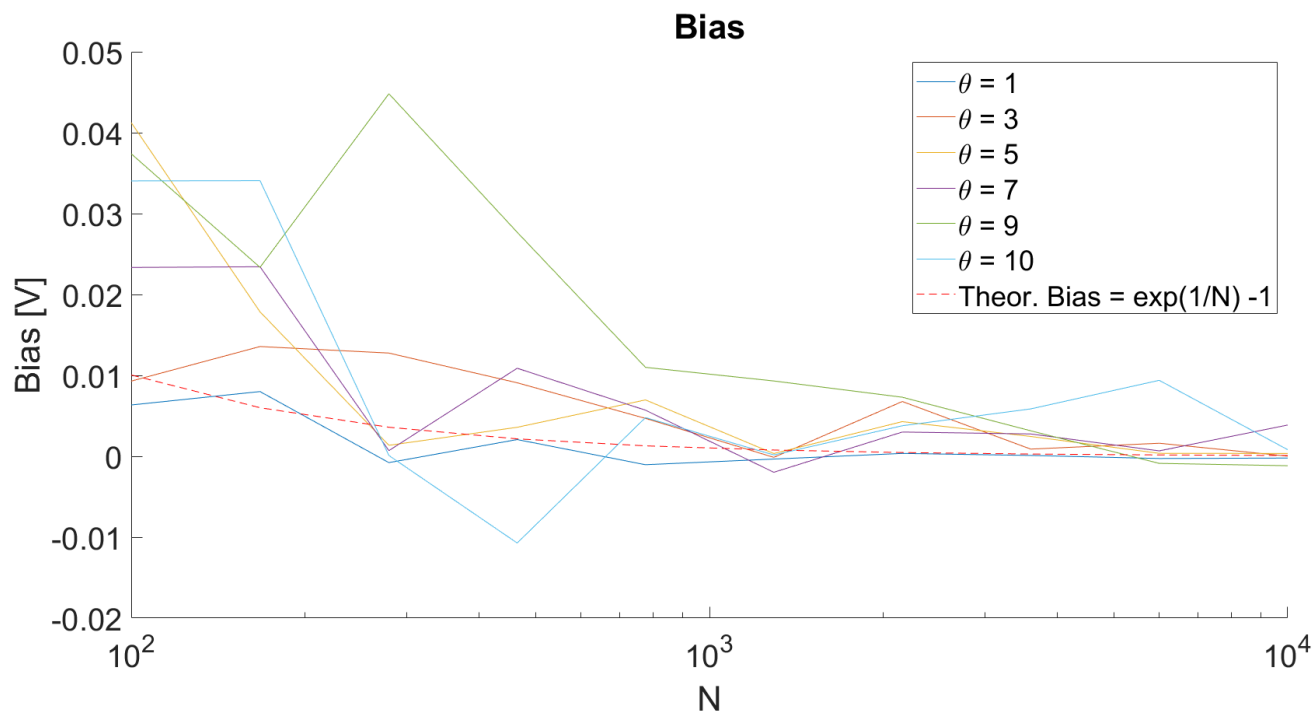
$$\text{MSE}(N, \theta) = \frac{1}{L} \sum_{i=1}^L \varepsilon_N^2(\mathbf{x}_i^{(N)}, \theta), \text{ respectively, and } L = 1000.$$

- i. Plot the bias, variance, MSE of the estimator and the Cramér-Rao bound (CRB) for estimation of θ , versus N for at least 5 different values of N . In each figure, include plots for 6 different values of θ in the interval $0.1 \leq \theta \leq 10$ where the N axis is logarithmic. In the figures of the variance, and MSE use a logarithmic scale for both axes.

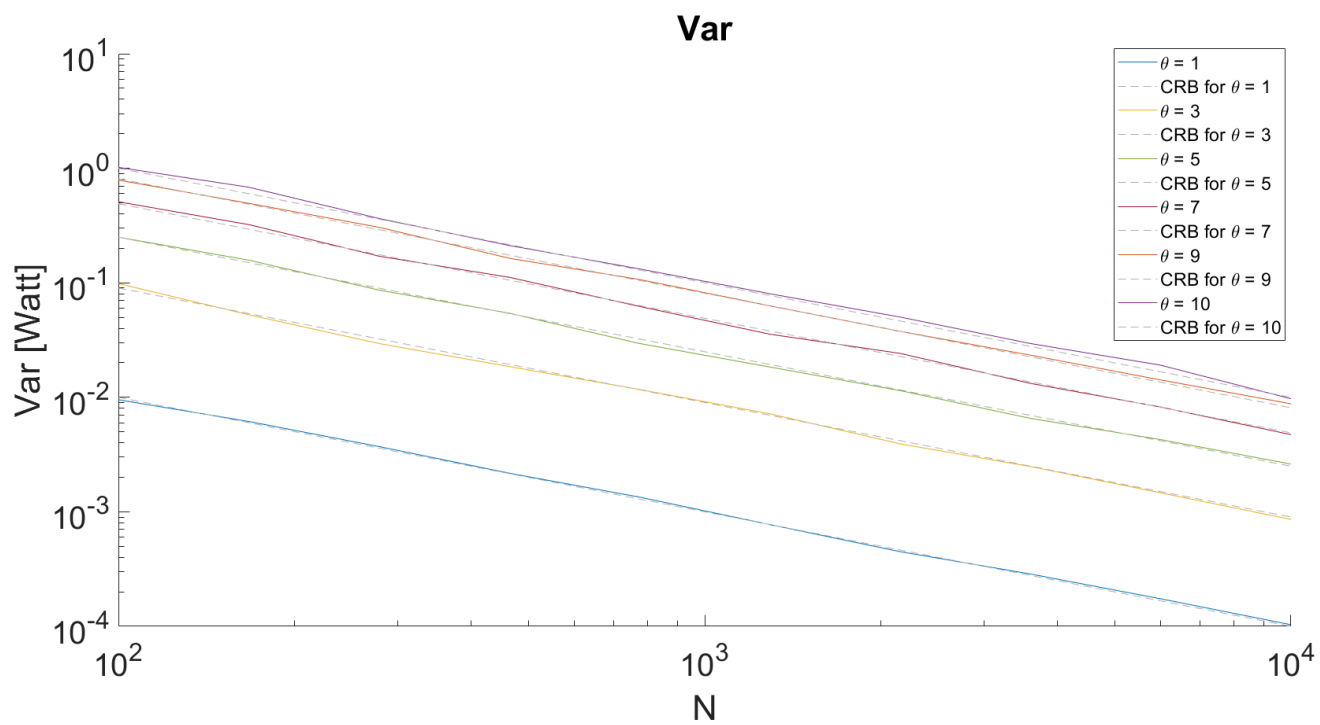
The results of said simulation are as follows

Plotted as a function of N :

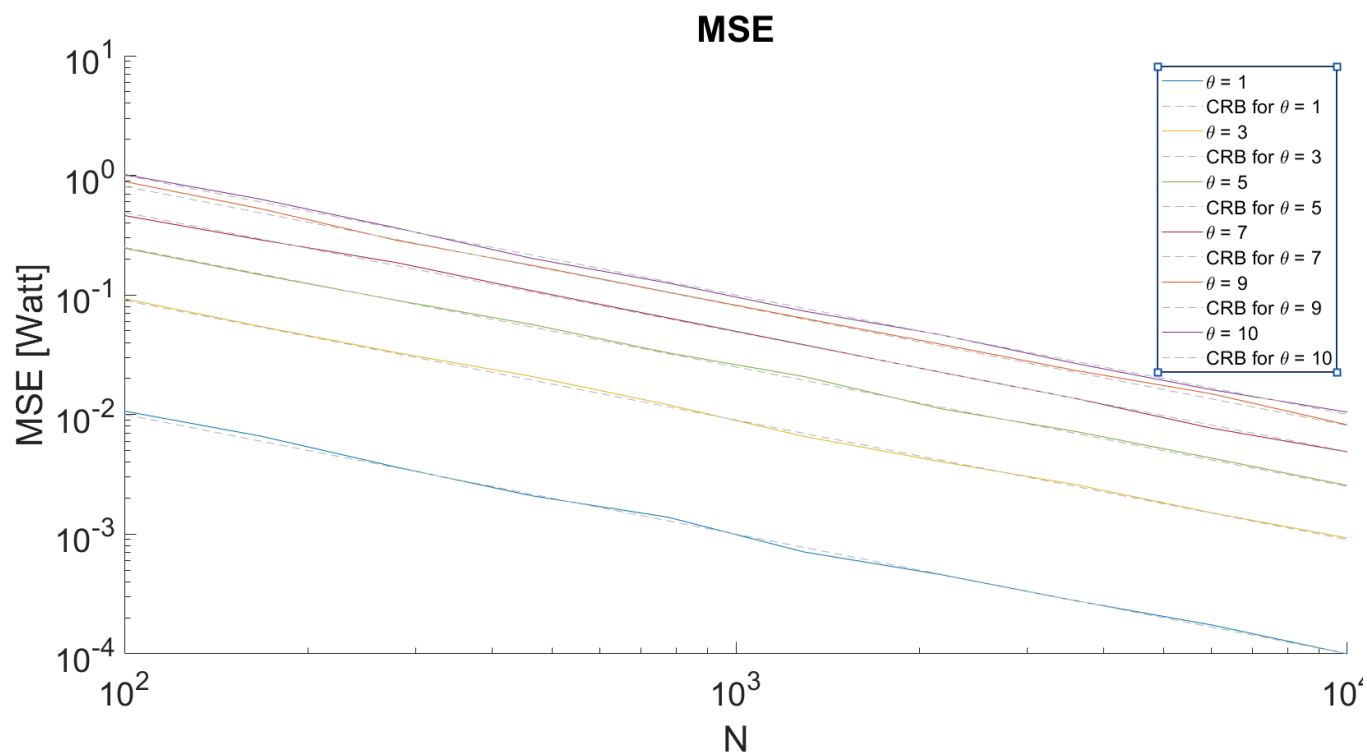
$\text{Bias}(N, \theta_i) :$



$Var(N, \theta_i) :$

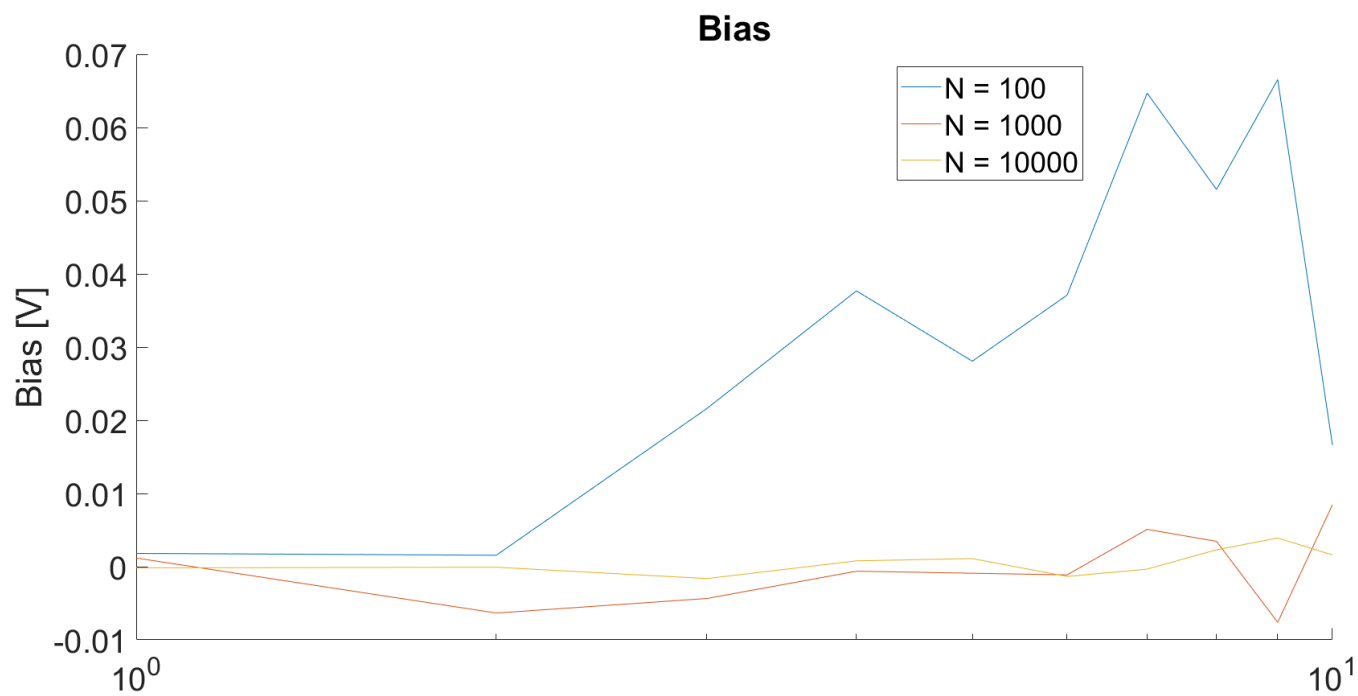


$MSE(N, \theta_i) :$

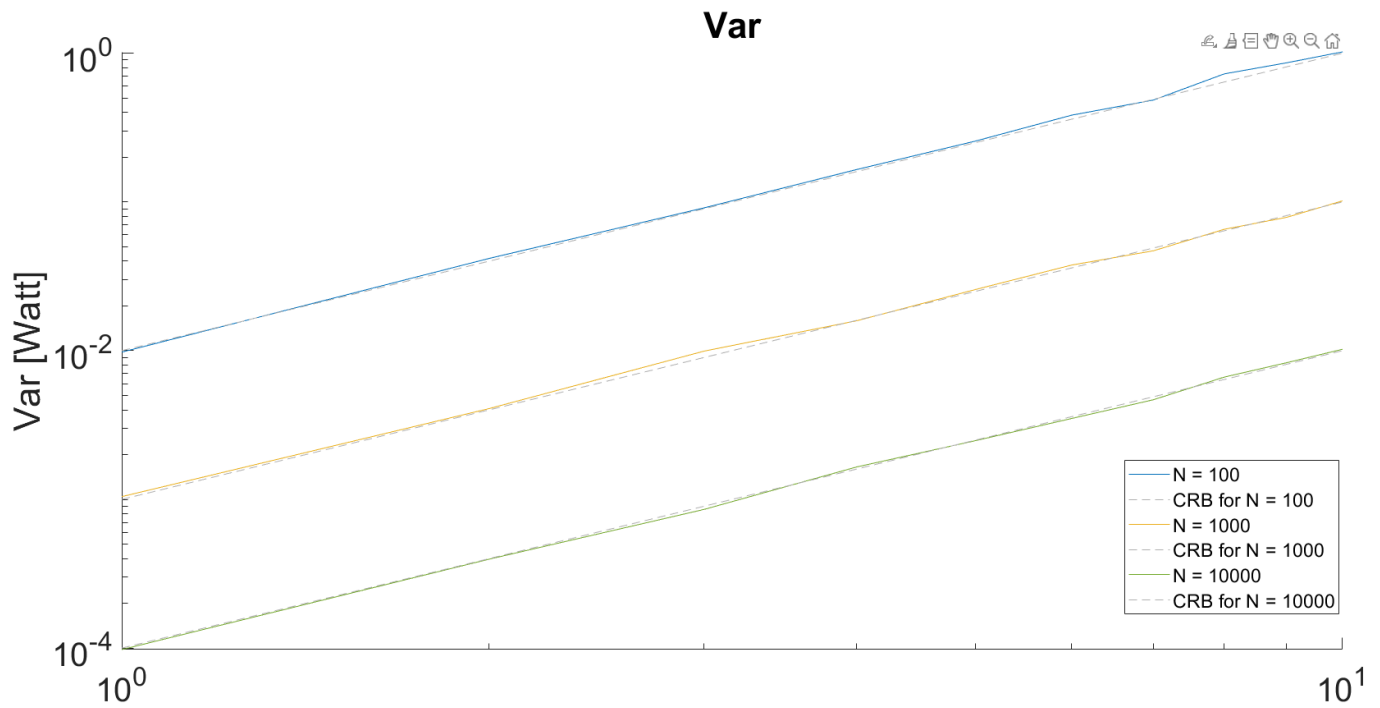


Plotted as a function of θ :

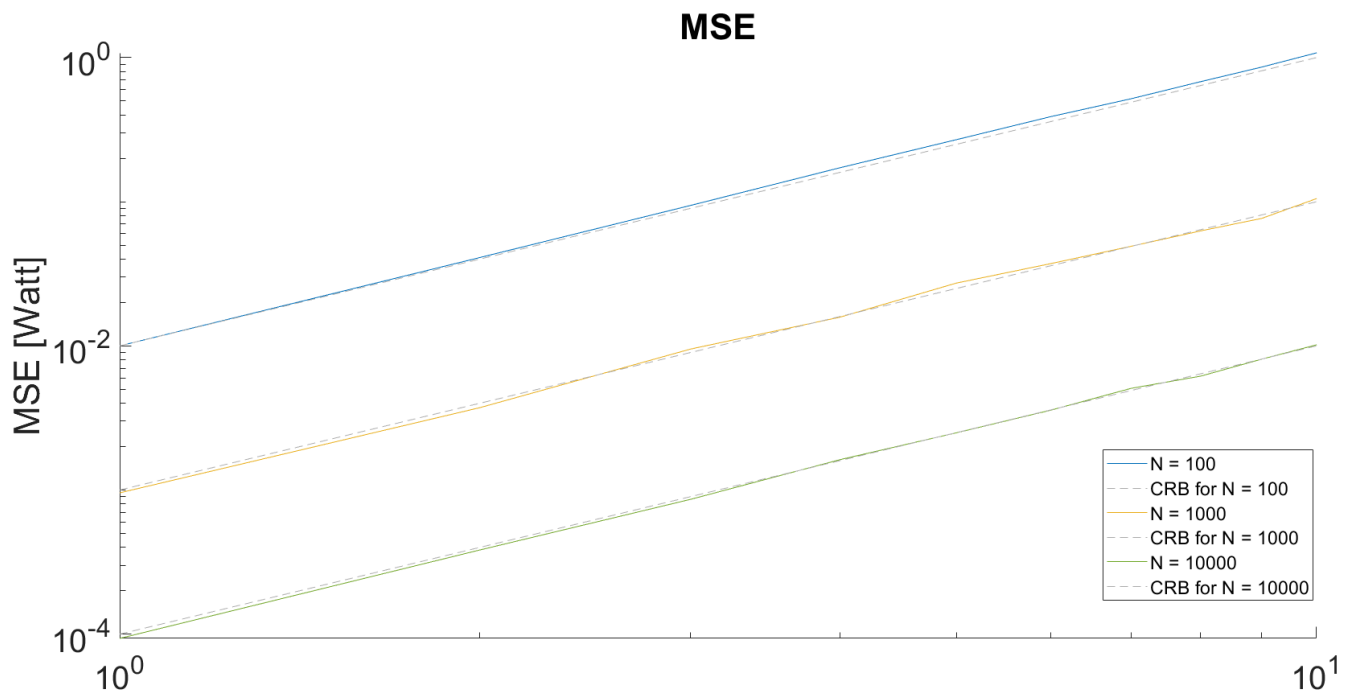
$Bias(N_i, \theta)$:



$Var(N_i, \theta)$:



$MSE(N_i, \theta) :$



We will calculate the CRLB from the definition :

$$Var(\hat{\theta}) \geq \mathcal{I}(\theta)^{-1} = CRLB$$

- We will derive the Fisher-Information from the definition. Since there are N i.i.d. samples, we can take the Fisher-Information of a single sample and multiply it by N :

$$\forall n, m \in \{1 \dots N\} : x_n \perp\!\!\!\perp x_m \implies \mathcal{I}(\theta) = N \cdot \mathbb{E} \left[\left(\frac{\partial \log f_{x_n|\theta}(x_n; \theta)}{\partial \theta} \right)^2 \right]$$

$$\log f_{x_n|\theta}(x_n; \theta) = \log \left[(2\pi)^{-\frac{1}{2}} \cdot \exp \left(-\frac{(x_n - \log \theta)^2}{2} \right) \right] = -\frac{1}{2} \log(2\pi) - \frac{(x_n - \log \theta)^2}{2}$$

$$\xrightarrow{\frac{\partial}{\partial \theta}} \cancel{\frac{1}{2}} (x_n - \log \theta) \cdot \cancel{2} \cdot (\cancel{-} \frac{1}{\theta}) = (x_n - \log \theta) \cdot \frac{1}{\theta}$$

$$N \cdot \mathbb{E} \left[\left(\frac{\partial \log f_{x_n|\theta}(x_n; \theta)}{\partial \theta} \right)^2 \right] = N \cdot \mathbb{E} \left[(x_n - \log \theta)^2 \cdot \frac{1}{\theta^2} \middle| \theta \right] = N \cdot \frac{1}{\theta^2} \cdot \mathbb{E}[(x_n - \log \theta)^2]$$

- We can notice that by definition, the expected value in question is the Variance of the conditional distribution $x_n|\theta$ which is given to be 1 :

$$\dots = N \cdot \frac{1}{\theta^2} \text{Var}(x_n|\theta) = N \cdot \frac{1}{\theta^2}$$

- We get the final expression for the Fisher-Information :

$$\implies \mathcal{I}(\theta) = \frac{N}{\theta^2}$$

✓ Result

And finally, we get the final expression for the CRLB :

$$\text{CRLB} = \mathcal{I}(\theta)^{-1} = \frac{\theta^2}{N}$$

Next, we want to check the condition for which the CRLB can be attained. As we proved in class, the conditional PDF must satisfy :

$$\frac{\partial \log f_{\underline{x}|\theta}(\underline{x}; \theta)}{\partial \theta} = c(\theta) \cdot ((x) - \theta)$$

- We will expand the terms for our problem (relying on earlier derivation...) to check if the condition holds :

$$\frac{\partial \log f_{\underline{x}|\theta}(\underline{x}; \theta)}{\partial \theta} = \dots = \underbrace{\frac{N}{\theta}}_{:=c(\theta)} \cdot \left[\underbrace{\frac{1}{N} \sum_{i=1}^N x_i}_{:= (x)} - \underbrace{\log \theta}_{\neq \theta} \right]$$

- There is no possible way to bring the $\log \theta$ term to the form of θ without breaking the definition for the rest of the terms...

✗ Result

We notice that we are not able to bring the score of the conditional distribution to the desired form and therefore we conclude that the CRLB is **unattainable** for our problem and in particular, to our estimator.

$$\implies \text{Var}(\hat{\theta}) > \mathcal{I}(\theta)^{-1} = \text{CRLB}$$

c) Efficiency

c. Is the proposed estimator efficient?

Since by definition an efficient estimator is an unbiased estimator (not asymptotically unbiased...) our proposed estimator is **not efficient**!

d) Efficiency

- d. Is the proposed estimator asymptotically efficient? Your answer should be based both on simulation results and on theoretical analysis.

...However, the estimator I proposed is only **Asymptotically Unbiased**. That means that for any finite sample vector, no matter how large it is, the estimator is biased and the attainability condition for the **CRLB** that was stated earlier does not apply to it.

For analysis purposes, We will derive the distribution of the estimator theoretically :

- First we want to evaluate the mean and bias of the estimator

$$\text{Bias}(\hat{\theta}, \theta) = \mathbb{E}[\hat{\theta} - \theta] = \mathbb{E}_{\underline{x}}[\hat{\theta}] - \theta$$

- To evaluate the following expected value we will use the MGF of a GWN vector :

$$\begin{aligned} \implies \mathbb{E}[\hat{\theta}] &= \mathbb{E}[e^{\frac{1}{N} \sum_{i=1}^N x_i}] = \prod_{i=1}^N \mathbb{E}[e^{\frac{x_i}{N}}] = \prod_{i=1}^N M_{x_i}(t) \Big|_{t=\frac{1}{N}} = \prod_{i=1}^N e^{(t \cdot \mu_i + \frac{1}{2} t^2 \cdot \sigma_i^2)} \Big|_{t=\frac{1}{N}} = \\ &= \prod_{i=1}^N e^{(\frac{1}{N} \cdot \log \theta + \frac{1}{2N^2} \cdot 1)} = e^{\log \theta} \cdot e^{\frac{1}{2N}} = \theta \cdot e^{\frac{1}{2N}} \end{aligned}$$

- We plug it back in the expression for the bias :

$$\implies \text{Bias}(\hat{\theta}, \theta) = \theta \cdot e^{\frac{1}{2N}} - \theta = \theta \cdot (e^{\frac{1}{2N}} - 1)$$

✓ **Result**

We get a final expression for the bias of the estimator and we notice that it is **Asymptotically Unbiased** :

$$\text{Bias}(\hat{\theta}, \theta) = \theta \cdot (e^{\frac{1}{2N}} - 1) \xrightarrow{N \rightarrow \infty} 0$$

- Now we want to find the variance of the estimator :

$$\text{Var}(\hat{\theta}) = \mathbb{E}[\hat{\theta}^2] - \mathbb{E}[\hat{\theta}]^2 = \dots$$

- We want to evaluate the second moment of the estimator so we will use the same MGF as before :

$$\begin{aligned} \mathbb{E}[\hat{\theta}^2] &= \mathbb{E}[(e^{\frac{1}{N} \sum_{i=1}^N x_i})^2] = \mathbb{E}[e^{\frac{2}{N} \sum_{i=1}^N x_i}] = \prod_{i=1}^N M_{x_i}(t) \Big|_{t=\frac{2}{N}} = \prod_{i=1}^N e^{(t \cdot \mu_i + \frac{1}{2} t^2 \cdot \sigma_i^2)} \Big|_{t=\frac{2}{N}} = \\ &= \prod_{i=1}^N e^{(\frac{2}{N} \cdot \log \theta + \frac{4}{2N^2} \cdot 1)} = e^{2 \log \theta} \cdot e^{\frac{4}{N}} = \theta^2 \cdot e^{\frac{2}{N}} \end{aligned}$$

- Since we've previously calculated the mean, we plug in all terms to get the expression variance of the estimator :

$$\implies \text{Var}(\hat{\theta}) = \theta^2 \cdot e^{\frac{2}{N}} - \left(\theta \cdot e^{\frac{1}{2N}} \right)^2 = \theta^2 \cdot (e^{\frac{2}{N}} - e^{\frac{1}{N}})$$

✓ Result

We get the expression variance of the estimator :

$$\text{Var}(\hat{\theta}) = \theta^2 \cdot (e^{\frac{2}{N}} - e^{\frac{1}{N}})$$

- We will use the **Taylor series of the exponent function** to evaluate the limit as the number of data samples goes to infinity :

$$e^x = 1 + x + o(x^2) \implies \text{Var}(\hat{\theta}) \approx \theta^2 \cdot \left(1 + \frac{2}{N} - \left(1 + \frac{1}{N} \right) \right) = \frac{1}{N} \cdot \theta^2$$

✓ Result

We can see that our estimator **attains the CRLB asymptotically** :

$$\text{Var}(\hat{\theta}) \propto \frac{\theta^2}{N}$$

⇒ From this we can also conclude that the estimator is **asymptotically efficient** !

It is visible from our plots that this projection holds since the bias always exist even though it is asymp. zero. The variance attains the CRB for each plot, matching our earlier projections. The MSE follows the same trend as the variance since the bias squared term goes to zero and the MSE and the variance align with each other.

Question 2

2. Let $x_1, \dots, x_N$ be statistically independent Gaussian random variables with mean μ and standard deviation σ .

We are given the statistical model of the data samples :

$$\underline{x} := \begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix} : \forall i \neq j \in \{1, \dots, n\} x_i \perp\!\!\!\perp x_j$$

$$\Rightarrow \underline{x} \sim \mathcal{N}(\mathbb{1} \cdot \mu, \sigma^2 \cdot \mathbb{I}_n)$$

a) + b) Find CRB and

- Find the CRB for estimation $[\mu, \sigma^2]^T$. Is there any coupling between the two parameters in the Fisher information matrix (FIM)?
- Does there exist an efficient estimator of $[\mu, \sigma^2]^T$? If yes, find the estimator.

We want to find the CRLB for the case of a GWN vector.

- We will find it based on the definition :

$$\underline{\theta} := [\mu, \sigma]^T \Rightarrow [\underline{\mathcal{I}}(\underline{\theta})]_{i,j} = \mathbb{E} \left[\frac{\partial \log f_{\underline{x}|\underline{\theta}}(\underline{x})}{\partial \theta_i} \cdot \frac{\partial \log f_{\underline{x}|\underline{\theta}}(\underline{x})}{\partial \theta_j} \right]$$

$$\log f_{\underline{x}|\underline{\theta}}(\underline{x}) = -\frac{n}{2} (\log(2\pi) + \log(\sigma^2)) - \sum_{i=1}^n \frac{(x_i - \mu)^2}{2\sigma^2}$$

$$\xrightarrow{\frac{d}{d\mu}} \sum_{i=1}^n \frac{(x_i - \mu)}{\sigma^2}$$

$$\xrightarrow{\frac{d}{d\sigma^2}} -\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2$$

- Calculating the off-diagonal terms :

$$\begin{aligned} [\mathcal{I}(\theta)]_{1,2} &= [\mathcal{I}(\theta)]_{2,1} = \mathbb{E} \left[\left[-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 \right] \cdot \sum_{i=1}^n (x_i - \mu) \right] = \\ &= \frac{1}{2\sigma^2} \mathbb{E} \left[n \sum_{i=1}^n \underbrace{(x_i - \mu)}_{x_i \sim \mathcal{N}(\mu, \sigma^2)} + \sum_{i=1}^n \frac{(x_i - \mu)^3}{\sigma^2} \right] = \frac{1}{2\sigma^4} \sum_{i=1}^n \mathbb{E} [(x_i - \mu)^3] = \end{aligned}$$

- To evaluate the expected value in question we will use the MGF of a joint-gaussian random vector :

$$\begin{aligned} \underline{z} := (\underline{x} - \mathbf{1} \cdot \mu) \sim \mathcal{N}(\underline{0}, \sigma^2 \cdot \mathbb{I}_n) &\implies \mathbb{E}[z_i^n] = \begin{cases} 0 & : \text{even} \\ \sigma^n \cdot (n-1)!! & : \text{odd} \end{cases} \\ \implies \mathbb{E}[(x_i - \mu)^3] = 0 &= [\mathcal{I}(\theta)]_{1,2} = [\mathcal{I}(\theta)]_{2,1} \end{aligned}$$

- Calculating the diagonal terms :

$$[\mathcal{I}(\theta)]_{1,1} = -\mathbb{E} \left[\frac{d}{d\mu} \left(\sum_{i=1}^n \frac{(x_i - \mu)}{\sigma^2} \right) \right] = \sum_{i=1}^n \frac{1}{\sigma^2} = \frac{n}{\sigma^2}$$

$$\begin{aligned} [\mathcal{I}(\theta)]_{2,2} &= -\mathbb{E} \left[\frac{d}{d(\sigma^2)} \left(-\frac{n}{2\sigma^2} + \frac{1}{2\sigma^4} \sum_{i=1}^n (x_i - \mu)^2 \right) \right] = -\mathbb{E} \left[\frac{n}{2\sigma^4} - \frac{1}{\sigma^6} \sum_{i=1}^n (x_i - \mu)^2 \right] = \\ &= -\frac{n}{2\sigma^4} + \frac{1}{\sigma^6} \sum_{i=1}^n \mathbb{E} [(x_i - \mu)^2] = -\frac{n}{2\sigma^4} + \frac{n \cdot \text{Var}(x_i)}{\sigma^6} = -\frac{n}{2\sigma^4} + \frac{n \cdot \sigma^2}{\sigma^6} = \frac{n}{2\sigma^4} \end{aligned}$$

- We get the FIM of the data samples :

$$\mathcal{I}(\theta) = \begin{bmatrix} \frac{n}{\sigma^2} & 0 \\ 0 & \frac{n}{2\sigma^4} \end{bmatrix}$$

✓ Result

We get the final expression for the CRB matrix :

$$\mathcal{I}(\theta)^{-1} = \begin{bmatrix} \frac{\sigma^2}{n} & 0 \\ 0 & \frac{2\sigma^4}{n} \end{bmatrix}$$

We can see that the off-diagonal terms are 0, then there is no coupling between the 2

which makes sense since the first and second moments of a distribution are independent of each other.

c) Variance Estimators bias :

c. Let $\hat{\sigma}^2 = \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2$, $\bar{x} = \frac{1}{N} \sum_{i=1}^N x_i$. Is $\hat{\sigma}^2$ an unbiased estimator of σ^2 ? If it is not, propose a proper correction of this estimator to obtain an unbiased estimator of σ^2 that is μ -independent.

- We will calculate the bias of the estimator :

$$\begin{aligned}\mathbb{E}[\hat{\sigma}^2] &= \mathbb{E} \left[\sum_{i=1}^N (x_i - \bar{x})^2 \right] = \sum_{i=1}^N \mathbb{E} [(x_i - \bar{x})^2] \\ \implies \mathbb{E} [(x_i - \bar{x})^2] &= \mathbb{E} [x_i^2] - 2 \cdot \mathbb{E} [x_i \cdot \bar{x}] + \mathbb{E} [\bar{x}^2]\end{aligned}$$

- We take advantage of the fact that : $\mathbb{E}[x^2] = \text{Var}(x) + \mathbb{E}[x]^2$

$$\forall k \neq i : \mathbb{E}[x_k \cdot x_i] = \begin{cases} \mu^2 & : i \neq k \\ \sigma^2 + \mu^2 & : i = k \end{cases}$$

$$\mathbb{E}[x_i^2] = \sigma^2 + \mu^2$$

$$\mathbb{E}[x_i \cdot \bar{x}] = \frac{1}{N} \sum_{k=1}^N \mathbb{E}[x_i x_k] = \frac{1}{N} (\sigma^2 + \mu^2 + (N-1) \cdot \mu^2) = \frac{1}{N} \sigma^2 + \mu^2$$

$$\mathbb{E}[\bar{x}^2] = \frac{1}{N^2} \sum_{k=1}^N \sum_{m=1}^N \mathbb{E}[x_k x_m] = \frac{1}{N^2} \cdot [N \cdot (\sigma^2 + \mu^2) + N(N-1) \cdot \mu^2] = \frac{1}{N} \cdot \sigma^2 + \mu^2$$

$$\implies \mathbb{E}[(x_i - \bar{x})^2] = \sigma^2 + \mu^2 - 2\left(\frac{1}{N} \cdot \sigma^2 + \mu^2\right) + \frac{1}{N} \cdot \sigma^2 + \mu^2 = \sigma^2 \left(1 - \frac{1}{N}\right)$$

$$\implies \mathbb{E}[\hat{\sigma}^2] = \frac{1}{N} \sum_{i=1}^N \sigma^2 \left(1 - \frac{1}{N}\right) = \frac{1}{N} \cdot N \cdot \sigma^2 \left(1 - \frac{1}{N}\right) = \sigma^2 \left(1 - \frac{1}{N}\right)$$

✓ Result

We see that the bias of the estimator is 0 only asymptotically, meaning that it is not unbiased:

$$\text{Bias}(\hat{\sigma}^2, \sigma^2) = \mathbb{E}[\hat{\sigma}^2 - \sigma^2] = \sigma^2 \left(1 - \frac{1}{N}\right) - \sigma^2 = \frac{\sigma^2}{N} \xrightarrow{N \rightarrow \infty} 0$$

We want to find the correction term make the estimator unbiased. We notice that we can multiply the original estimator by a function of the number of samples and remove the bias :

- We propose the following estimator $\tilde{\sigma}^2$:

$$\tilde{\sigma}^2 := \frac{1}{1 - \frac{1}{N}} \hat{\sigma}^2 = \frac{1}{1 - \frac{1}{N}} \frac{1}{N} \sum_{i=1}^N (x_i - \bar{x})^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2$$

- We notice that the expected value of the estimator comes out to be the target parameter :

$$\implies \mathbb{E}[\tilde{\sigma}^2] = \frac{1}{1 - \frac{1}{N}} \mathbb{E}[\hat{\sigma}^2] = \cancel{\frac{1}{1 - \frac{1}{N}}} \cdot \sigma^2 \cancel{\left(1 - \frac{1}{N}\right)} = \sigma^2$$

✓ Result

We see that the new estimator we propose is unbiased :

$$\tilde{\sigma}^2 = \frac{1}{N-1} \sum_{i=1}^N (x_i - \bar{x})^2 \implies \mathbb{E}[\tilde{\sigma}^2] = \sigma^2$$

d) Variance of Proposed Estimator

- d. What is the variance of the estimator you obtained after the correction of $\hat{\sigma}^2$?
Does this estimator achieve the corresponding CRB from Section a?

Note: You can use the fact that $\{x_i - \bar{x}\}_{i=1}^N$ are jointly Gaussian random variables with zero mean. In addition, you can use the following identity for zero-mean jointly Gaussian random variables:

$$\mathbb{E}[XYZW] = \mathbb{E}[XY]\mathbb{E}[ZW] + \mathbb{E}[XZ]\mathbb{E}[YW] + \mathbb{E}[XW]\mathbb{E}[YZ] .$$

We will calculate the variance of the estimator and check if it achieves the CRB.

$$\text{Var}(\tilde{\sigma}^2) = \mathbb{E}[(\tilde{\sigma}^2)^2] - \mathbb{E}[\tilde{\sigma}^2]^2$$

$$\mathbb{E}[(\tilde{\sigma}^2)^2] = \mathbb{E} \left[\frac{1}{(N-1)^2} \sum_{i=1}^N \sum_{k=1}^N (x_i - \bar{x})^2 \cdot (x_k - \bar{x})^2 \right] = \frac{1}{(N-1)^2} \sum_{i=1}^N \sum_{k=1}^N \mathbb{E} [(x_i - \bar{x})^2 \cdot (x_k - \bar{x})^2]$$

- We would like to evaluate the expected value $\mathbb{E} [(x_i - \bar{x})^2 \cdot (x_k - \bar{x})^2]$ inside the sum. For that we will use the fact that the R.V.s $\{x_i - \bar{x}\}_{i=1}^N$ are jointly Gaussian with zero mean. Therefore, we can denote $z_k := x_k - \bar{x}$ and expand the expected value term as follows :

$$\mathbb{E} [(x_i - \bar{x})^2 \cdot (x_k - \bar{x})^2] = \mathbb{E}[z_i \cdot z_i \cdot z_k \cdot z_k] =$$

$$\begin{aligned}
&= \mathbb{E}[z_i \cdot z_i] \cdot \mathbb{E}[z_k \cdot z_k] + \mathbb{E}[z_i \cdot z_k] \cdot \mathbb{E}[z_k \cdot z_i] + \mathbb{E}[z_i \cdot z_k] \cdot \mathbb{E}[z_k \cdot z_i] = \mathbb{E}[z_i^2]^2 + 2 \cdot \mathbb{E}[z_i \cdot z_k]^2 \\
&\implies \mathbb{E}[(x_i - \bar{x})^2 \cdot (x_k - \bar{x})^2] = \mathbb{E}[z_i^2]^2 + 2 \cdot \mathbb{E}[z_i \cdot z_k]^2
\end{aligned}$$

- We have calculated the first term in subsection c) :

$$\begin{aligned}
\mathbb{E}[z_i^2] &= \mathbb{E}[(x_i - \bar{x})^2] = \sigma^2 \left(1 - \frac{1}{N}\right) \\
\implies \mathbb{E}[z_i^2]^2 &= \sigma^4 \left(1 - \frac{1}{N}\right)^2
\end{aligned}$$

- We want to find $\mathbb{E}[z_i \cdot z_k]$ to evaluate the quantity above.

$$\begin{aligned}
\mathbb{E}[z_i \cdot z_k] &= \underbrace{\mathbb{E}[x_i \cdot x_k]}_{\dots = \mu^2 + \delta(i-k) \cdot \sigma^2} - \mathbb{E}[x_i \cdot \bar{x}] - \mathbb{E}[x_k \cdot \bar{x}] + \mathbb{E}[\bar{x}^2] = \\
&= \mu^2 + \delta(i-k) \cdot \sigma^2 - 2 \cdot \left(\frac{1}{N} \sigma^2 + \mu^2\right) + \left(\frac{1}{N} \sigma^2 + \mu^2\right) = \sigma^2 \cdot \delta(i-k) - \frac{\sigma^2}{N}
\end{aligned}$$

- We plug it back in the previous term :

$$\begin{aligned}
&\implies \mathbb{E}[z_i^2]^2 + 2 \cdot \mathbb{E}[z_i \cdot z_k]^2 = \sigma^4 \left(1 - \frac{1}{N}\right)^2 + 2 \cdot \left(\sigma^2 \cdot \delta(i-k) - \frac{\sigma^2}{N}\right)^2 = \\
&= \sigma^4 \left[\left(1 - \frac{1}{N}\right)^2 + 2 \cdot \left(\delta(i-k) - \frac{1}{N}\right)^2 \right] = \sigma^4 \left[1 - \frac{2}{N} + \frac{1}{N^2} + 2 \cdot \left(\delta(i-k) - \delta(i-k) \frac{2}{N} + \frac{1}{N^2}\right)^2 \right] \\
&\implies \mathbb{E}[(x_i - \bar{x})^2 (x_k - \bar{x})^2] = \sigma^4 \left[1 + 2\delta(i-k) - (2\delta(i-k) + 1) \cdot \frac{2}{N} + \frac{3}{N^2} \right]
\end{aligned}$$

- We count the amount of time that $i = k$ and the rest of the times and substitute accordingly :

$$\begin{aligned}
&\frac{1}{(N-1)^2} \sum_{i=1}^N \sum_{k=1}^N \mathbb{E}[(x_i - \bar{x})^2 \cdot (x_k - \bar{x})^2] = \\
&= \frac{1}{(N-1)^2} \sum_{i=1}^N \sum_{k=1}^N \sigma^4 \left[1 + 2\delta(i-k) - (2\delta(i-k) + 1) \cdot \frac{2}{N} + \frac{3}{N^2} \right] = \\
&= \frac{\sigma^4}{(N-1)^2} \cdot \left[N \cdot \left(3 - \frac{6}{N} + \frac{3}{N^2}\right) + N(N-1) \left(1 - \frac{2}{N} + \frac{3}{N^2}\right) \right] = \\
&= \frac{\sigma^4}{(N-1)^2} \cdot \left[3N - 6 + \frac{3}{N} + N(N-1) - 2(N-1) + \frac{3(N-1)}{N} \right] = \\
&\implies \mathbb{E}[(\tilde{\sigma}^2)^2] = \frac{\sigma^4}{(N-1)^2} (N^2 - 1) = \sigma^4 \frac{N+1}{N-1}
\end{aligned}$$

✓ Result

Since the estimator we proposed is unbiased we know that : $\mathbb{E}[\tilde{\sigma}^2] = \sigma^2$

And therefore, we get the expression for the variance of the estimator :

$$\begin{aligned} Var(\tilde{\sigma}^2) &= \mathbb{E}[(\tilde{\sigma}^2)^2] - \mathbb{E}[\tilde{\sigma}^2]^2 = \frac{\sigma^4}{(N-1)^2} (N^2 - 1) = \sigma^4 \frac{N+1}{N-1} - (\sigma^2)^2 \\ \implies Var(\tilde{\sigma}^2) &= \frac{2\sigma^4}{N-1} \neq \frac{2\sigma^4}{N} = CRLB \end{aligned}$$

✗ **Not Efficient**

We can see that for any finite number of samples the CRLB is not attained!

e) $N \rightarrow \infty$:

We repeat d) for the limit as N goes to infinity.

- Consistency :

$$\lim_{N \rightarrow \infty} Var(\tilde{\sigma}^2) = \lim_{N \rightarrow \infty} \frac{2\sigma^4}{N-1} = 0$$

- We can see that the estimator is consistent
- Efficiency :

$$\lim_{N \rightarrow \infty} \frac{Var(\tilde{\sigma}^2)}{CRLB} = \lim_{N \rightarrow \infty} \frac{\frac{2\sigma^4}{N-1}}{\frac{2\sigma^4}{N}} = \lim_{N \rightarrow \infty} \frac{N}{N-1} = 1$$

- Therefore, the proposed estimator is **Asymptotically Efficient** !

Question 3

Note 1: Assume that the matrix $\mathbf{Q}$ is invertible and that the partial derivatives, up to order n , of $f(\mathbf{x}; \theta)$ with respect to θ , exist and are absolutely integrable.

Note 2: Assume that the regularity condition of Cramér-Rao bound is satisfied.

a) Bhattacharyya Bound

- a. Prove that for any unbiased estimator of θ (denoted by $\hat{\theta}$), the following

equality is satisfied:
$$\mathbb{E} \left\{ \left(\hat{\theta} - \theta \right) \frac{1}{f(\mathbf{x}; \theta)} \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} \right\} = \begin{cases} 1 & i=1 \\ 0 & i \geq 2 \end{cases}.$$

We expand the terms by definition :

$$\begin{aligned}\mathbb{E} \left[(\hat{\theta} - \theta) \frac{1}{f(\mathbf{x}; \theta)} \cdot \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} \right] &= \int_{\mathbb{R}^N} \left[(\hat{\theta} - \theta) \frac{1}{\cancel{f(\mathbf{x}; \theta)}} \cdot \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} \right] \cancel{f(\mathbf{x}; \theta)} d\mathbf{x} = \\ &= \int_{\mathbb{R}^N} (\hat{\theta} - \theta) \cdot \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} d\mathbf{x} = \int_{\mathbb{R}^N} \hat{\theta} \cdot \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} d\mathbf{x} - \int_{\mathbb{R}^N} \theta \cdot \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} d\mathbf{x} =\end{aligned}$$

- Under regularity condition we can switch between integration and differentiation and since θ isn't a function of $\mathbf{x}$ and $\hat{\theta}$ is, we can take it out the integral :

$$= \frac{\partial^i}{\partial \theta^i} \int_{\mathbb{R}^N} \hat{\theta} \cdot f(\mathbf{x}; \theta) d\mathbf{x} - \theta \cdot \frac{\partial^i}{\partial \theta^i} \int_{\mathbb{R}^N} f(\mathbf{x}; \theta) d\mathbf{x} =$$

- We notice the following :

1. Since the estimator is given to be unbiased we get :

$$\int_{\mathbb{R}^N} \hat{\theta} \cdot f(\mathbf{x}; \theta) d\mathbf{x} = \mathbb{E}[\hat{\theta}] = \theta$$

2. Since the PDF $f(\mathbf{x}; \theta)$ is a valid PDF, the integral over all of its domain is 1 :

$$\int_{\mathbb{R}^N} f(\mathbf{x}; \theta) d\mathbf{x} = 1$$

- So basically the expression we are evaluating is :

$$\dots = \frac{\partial^i}{\partial \theta^i}(\theta) - \underbrace{\frac{\partial^i}{\partial \theta^i}(1)}_{=0 : \forall i \geq 1} = \frac{\partial^i}{\partial \theta^i}(\theta) = \begin{cases} 1 & : i = 1 \\ 0 & : i \geq 2 \end{cases}$$

✓ Result

We have proven the identity :

$$\mathbb{E} \left[(\hat{\theta} - \theta) \frac{1}{f(\mathbf{x}; \theta)} \cdot \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} \right] = \begin{cases} 1 & : i = 1 \\ 0 & : i \geq 2 \end{cases}$$

b) Finding Lower bound

- b. Use the Cauchy-Schwarz inequality to find a lower bound for $\mathbb{E}\left[(\hat{\theta} - \theta)^2\right]$.

Express the bound by the $n \times n$ matrix $\mathbf{Q}$, whose elements are given by

$$Q_{ij} = \mathbb{E}\left\{\frac{1}{f^2(\mathbf{x}; \theta)} \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} \frac{\partial^j f(\mathbf{x}; \theta)}{\partial \theta^j}\right\}, \quad i, j = 1, \dots, n.$$

We will use the Cauchy-Schwarz inequality to find the lower bound of $\mathbb{E}\left[(\hat{\theta} - \theta)^2\right]$.

- We are asked to define the bound using the matrix $\mathbf{Q} \in \mathbb{R}^n$ while :

$$Q_{ij} = \mathbb{E}\left[\frac{1}{f^2(\mathbf{x}; \theta)} \cdot \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i} \cdot \frac{\partial^j f(\mathbf{x}; \theta)}{\partial \theta^j}\right] : i, j \in \{1, 2, \dots, n\}$$

- We will define $\mathbf{z} = [z_1, z_2, \dots, z_n]^T$ to be a random vector of the form :

$$z_i = \frac{1}{f(\mathbf{x}; \theta)} \frac{\partial^i f(\mathbf{x}; \theta)}{\partial \theta^i}$$

- From the definition of the matrix $\mathbf{Q}$ we notice that :

$$\mathbf{Q} = \mathbb{E}[\mathbf{z}\mathbf{z}^T]$$

- We will calculate the following cross-covariance:

$$\mathbf{v} = \mathbb{E}[(\hat{\theta} - \theta)\mathbf{z}]$$

- If we plug in the full term we get the result from a) :

$$\mathbf{v}_i = \begin{cases} 1 & : i = 1 \\ 0 & : i \geq 2 \end{cases} \implies \mathbf{v} = [1, 0, \dots, 0]^T$$

- We can take the covariance matrix of the random vector $\begin{bmatrix} \hat{\theta} - \theta \\ \mathbf{z} \end{bmatrix}$ and notice that, by definition, it must be a semi-positive matrix :

$$\mathbb{E}\left[\begin{bmatrix} \hat{\theta} - \theta \\ \mathbf{z} \end{bmatrix} \begin{bmatrix} \hat{\theta} - \theta & \mathbf{z}^T \end{bmatrix}\right] = \begin{bmatrix} \mathbb{E}[(\hat{\theta} - \theta)^2] & \mathbf{v}^T \\ \mathbf{v} & \mathbf{Q} \end{bmatrix} \succeq 0$$

- We can apply the **Schur complement** of the covariance matrix and get the following inequality :

$$\mathbb{E}[(\hat{\theta} - \theta)^2] - \mathbf{v}^T \mathbf{Q}^{-1} \mathbf{v} \geq 0 \implies \mathbb{E}[(\hat{\theta} - \theta)^2] \geq \mathbf{v}^T \mathbf{Q}^{-1} \mathbf{v}$$

- Since the vector $v = [1, 0, \dots, 0]^T$ when multiplying the terms we actually extract the top left entry of the matrix Q^{-1}

✓ Result

We get the final inequality of the target expected value that defines the Bhattacharyya bound:

$$\mathbb{E}[(\hat{\theta} - \theta)^2] \geq [\mathbf{Q}^{-1}]_{11}$$

c) Comparison to the CRLB

- c. Prove that the resulting bound in Section b is larger than or equal to the Cramér-Rao bound (Hint: Check the case $n = 1$).

We will prove by induction that : Bhattacharyya Bound $\geq$ Cramer-Rao Bound

1. Base : $n = 1$

For the base case, Q and $I(\theta)$ are scalar matrices :

$$Q_{11} = \mathbb{E} \left[\left(\frac{1}{f(\mathbf{x}; \theta)} \frac{\partial f(\mathbf{x}; \theta)}{\partial \theta} \right)^2 \right] = \mathbb{E} \left[\left(\frac{\partial \ln f(\mathbf{x}; \theta)}{\partial \theta} \right)^2 \right] = I(\theta)$$

We can see that the 2 matrices are equal by definition and therefore, so are their inverses :

$$\mathbb{E}[(\hat{\theta} - \theta)^2] \geq Q_{11}^{-1} = \frac{1}{I(\theta)} = \text{CRB}$$

$\implies$ proven for the case of $n = 1$!

2. Step : $n > 1$

We can decompose the matrix Q of dimension n as follows :

$$\mathbf{Q}_n = \begin{bmatrix} Q_{11} & \mathbf{R}^T \\ \mathbf{R} & \mathbf{S} \end{bmatrix}$$

Where Q_{11} was previously evaluated, $\mathbf{R}, \mathbf{R}^T \in \mathbb{R}^n$ is some vector (covariance is symmetric...) and $\mathbf{S} \in \mathbb{R}^{(N-1) \times (N-1)}$ is some symmetric matrix.

From a formula we derived in Assignment 1 for the inverse of a matrix of such structure we get:

$$[\mathbf{Q}_n^{-1}]_{11} = (Q_{11} - \mathbf{R}^T \mathbf{S}^{-1} \mathbf{R})^{-1}$$

We have already established that $\mathbf{S}$ is positive-definite so we can safely claim that :

$$\mathbf{R}^T \mathbf{S}^{-1} \mathbf{R} \geq 0 : \forall \mathbf{R} \in \mathbb{R}^N$$

Therefore the following inequality must hold for every number, specifically for Q_{11} :

$$Q_{11} - \mathbf{R}^T \mathbf{S}^{-1} \mathbf{R} \geq Q_{11}$$

We take the reciprocal of both sides, and compare with what we got before :

$$\implies \mathbb{E}[(\hat{\theta} - \theta)^2] \geq [\mathbf{Q}_n^{-1}]_{11} = (Q_{11} - \mathbf{R}^T \mathbf{S}^{-1} \mathbf{R})^{-1} \leq Q_{11}^{-1} = I(\theta)^{-1}$$

✓ Result

We have shown that for any $n \in \mathbb{N}$ the inequality between the bounds holds :

$$\begin{aligned} \implies [\mathbf{Q}_n^{-1}]_{11} &\geq Q_{11}^{-1} \\ \implies \text{Bhattacharyya Bound} &\geq \text{Cramer-Rao Bound} \end{aligned}$$

Question 4

4. The Cramér-Rao bound can be derived as follows. Define the vector

$$\mathbf{z} = \begin{pmatrix} \hat{\theta} - \theta \\ \frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta} \end{pmatrix}$$

Note: Assume that the regularity condition of Cramér-Rao bound is satisfied.

a) Verify Claim

a. Verify that for any unbiased estimator $\hat{\theta}$, $\mathbb{E}(\mathbf{z}) = \mathbf{0}$.

We will take the expectation of the random vector.

- We plug in the terms :

$$\mathbb{E}[\mathbf{z}] = \begin{pmatrix} \mathbb{E}[\hat{\theta} - \theta] \\ \mathbb{E}\left[\frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta}\right] \end{pmatrix}$$

- We are given that said estimator is unbiased, therefore :

$$\mathbb{E}[\hat{\theta} - \theta] = 0$$

- We expand the second entry by definition and we expand the differentiation of the logarithm :

$$\mathbb{E} \left[\frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta} \right] = \int_{\mathbb{R}^N} \frac{1}{f(\mathbf{x}; \theta)} \frac{\partial f(\mathbf{x}; \theta)}{\partial \theta} \cdot \cancel{f(\mathbf{x}; \theta)} d\mathbf{x} = \frac{\partial}{\partial \theta} \int_{\mathbb{R}^N} f(\mathbf{x}; \theta) d\mathbf{x} = \frac{\partial}{\partial \theta} (1) = 0$$

✓ Result

We have shown the target identity :

$$\mathbb{E}[\mathbf{z}] = \begin{pmatrix} \mathbb{E}[\hat{\theta} - \theta] \\ \mathbb{E}[\frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta}] \end{pmatrix} = \mathbf{0}$$

b) Covariance

b. Derive an expression for the covariance matrix of $\mathbf{z}$, $\Lambda = \mathbb{E}\{\mathbf{z} \mathbf{z}^T\}$.

We will derive the expression by definition :

$$\Lambda = \mathbb{E}[\mathbf{z} \mathbf{z}^T] = \begin{bmatrix} \mathbb{E}[(\hat{\theta} - \theta)^2] & \mathbb{E}\left[(\hat{\theta} - \theta) \cdot \frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta}\right] \\ \mathbb{E}\left[(\hat{\theta} - \theta) \cdot \frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta}\right] & \mathbb{E}\left[\left(\frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta}\right)^2\right] \end{bmatrix}$$

1. Λ_{11} :

Since the estimator is unbiased we can safely claim that

$$\mathbb{E}[(\hat{\theta} - \theta)^2] = \text{Var}(\hat{\theta}) + \cancel{\text{Bias}(\theta)} = \text{Var}(\hat{\theta})$$

2. Λ_{22} :

It is quite clear that the target expected value is by definition the fisher information of the estimation of θ from the samples $\mathbf{x}$:

$$\mathbb{E}\left[\left(\frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta}\right)^2\right] = I(\theta)$$

3. $\Lambda_{12}, \Lambda_{21}$:

In [Question 2.a\)](#), we have proven that for the first order derivative is 1 :

$$\mathbb{E}\left[(\hat{\theta} - \theta) \cdot \frac{\partial^i \log f(\mathbf{x}; \theta)}{\partial^i \theta}\right] \Big|_{i=1} = 1$$

✓ Result

We get the final derivation of the covariance matrix of $\mathbf{z}$:

$$\Lambda = \begin{bmatrix} \text{Var}(\hat{\theta}) & 1 \\ 1 & I(\theta) \end{bmatrix}$$

c) Proving the CRLB

- c. Use the fact that Λ is a positive semidefinite matrix to obtain the Cramér-Rao inequality.

Since $\Lambda \succeq 0$ can apply its **Schur Complement** to get the following inequality :

$$\Lambda = \begin{bmatrix} \text{Var}(\hat{\theta}) & 1 \\ 1 & I(\theta) \end{bmatrix} \succeq 0 \implies \text{Var}(\hat{\theta}) - (1)^T \cdot I(\theta)^{-1} \cdot (1) \geq 0$$

✓ Result

We get the original CRLB of the Variance for estimating θ :

$$\implies \text{Var}(\hat{\theta}) \geq \frac{1}{I(\theta)}$$

Question 5

5. Let $w_1, \dots, w_N$ be i.i.d. random variables with probability density function (PDF)

$$f_{w_n}(w) = \frac{1}{2} \exp(-|w|). \text{ We are interested to estimate } \theta \text{ from the observations}$$

$$x_n = \theta + w_n, \quad n = 1, \dots, N.$$

a) Confirm Regularity

- a. Confirm that the regularity condition of Cramér-Rao bound is satisfied.

We want to show the following :

$$\mathbb{E} \left[\frac{\partial \log f(\mathbf{x}; \theta)}{\partial \theta} \right] = 0$$

- First of all, we want to find the conditional distribution of $\mathbf{x}|\theta$.